GG-S-321D
March 9, 1972
SUPERSEDING
Fed. Spec. GG-S-321C
July 9, 1963

FEDERAL SPECIFICATION

SHIELDS, ERASING, DRAFTSMEN'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one type of metal draftsman erasing shield.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-766 - Steel Plates, Sheets, and Strip, Corrosion Resisting.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Method.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 7510

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The shields shall be manufactured from corrosion-resisting steel sheet conforming to QQ-S-766, Class 301, 302, 304, or 305; Condition, A; Temper, optional; Finish, number 7.

3.1.1 Thickness. The thickness of the steel sheets shall be not less than 0.0045 inch and not more than 0.007 inch.

3.2 Design and construction. The shield shall be made of material as specified in 3.1, and shall be of a thickness as specified in 3.1.1. The width and length of the shield shall be as shown in figure 1, excluding rounded corners. The contour of the openings, and the number of the openings shall be as shown in figure 1, except that additional openings may be provided. The arrangements of the openings shall conform to the manufacturer's commercial practice. The corners of the shield shall be rounded.

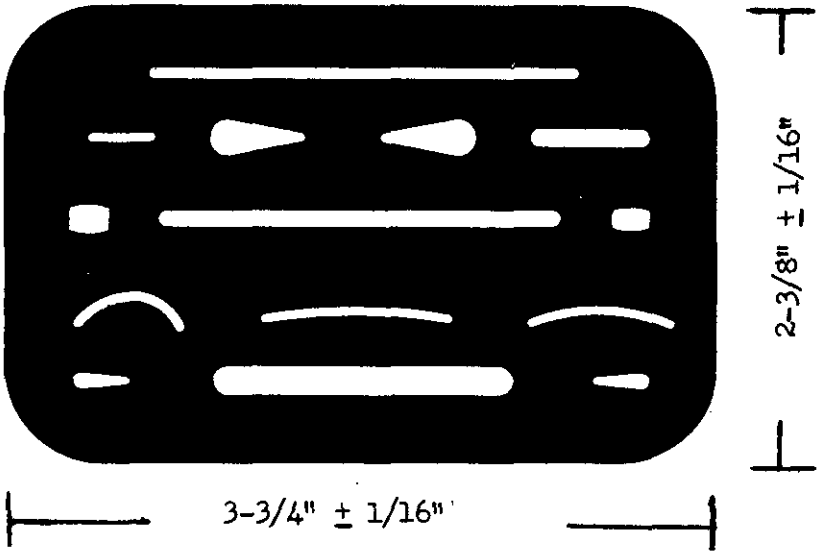


FIGURE 1

3.3 Finish. Both sides of the shield shall be given a high polish (see 3.1).

3.4 Flexibility (recoil). The shield shall recoil without assistance to its original plane shape within a period of 10 minutes, when subjected to the test specified in 4.4.1.1.

3.5 Identification marking. Each envelope of shields shall be legibly marked with the name or trademark of the manufacturer.

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3.6 Workmanship. The shields shall be free from defects or imperfections which adversely affect appearance, impair serviceability, or constitute a hazard to the users. The occurrence of defects shall not exceed the applicable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Size of lot. All draftsman shields presented for inspection at one time shall be considered a lot for purpose of examination and tests.

4.2.2 Sampling for examination and tests. Sampling for examination and tests shall be performed in accordance with MIL-STD-105, except where otherwise specified.

4.2.3 Sampling for visual and dimensional inspection. Random sample envelopes of shields shall be taken from each lot in accordance with MIL-STD-105. A sample unit shall be one envelope of shields, except that three (3) shields, taken at random, shall be examined for defects listed in table I. The Inspection Level shall be S-2, and the Acceptable Quality Level (AQL) shall be 2.5 percent defective.

4.2.4 Sampling for tests. For each test, random sample envelopes of shields shall be taken from each lot in accordance with MIL-STD-105. A sample unit shall be one envelope of shields, except that two shields, taken at random, shall be tested as specified in 4.4. The Inspection Level shall be S-3, and the AQL shall be 4.0 percent defective.

4.2.5 Sampling for preparation for delivery requirements. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The Inspection Level shall be S-2, with an AQL of 4.0 defects per hundred units.

4.3 Examination.

4.3.1 Visual and dimensional examination. Samples taken in accordance with 4.2.3 shall be examined for defects listed in table I. Any shield having one or more defects shall be rejected. If the number of defective units in the sample exceeds the acceptance number for that sample, the entire lot represented by the sample shall be rejected.

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TABLE I. Classification of defects

Examine	Defects
Material	Not as specified (see 3.1 and 3.1.1).
Design	Not as specified (see 3.2).
Finish	Not as specified (see 3.3).
Identification marking	Not as specified (see 3.5).
Workmanship	Not in accordance with 3.6.

4.3.2 Examination of preparation for delivery defects. Samples selected in accordance with 4.2.5 shall be examined for defects listed in table II to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with.

TABLE II. Classification of preparation for delivery

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect, illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents (exterior container)	Number per container is more or less than required; net weight exceeds requirements.

4.4 Tests.

4.4.1 End item tests. Samples selected in accordance with 4.2.4 shall be subjected to the following tests. Any shield in the sample failing the test shall be considered defective. If the number of defective shields exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4.1.1 Flexibility test. The sample shall be bent lengthwise around a right circular cylinder 2 inches in diameter, and shall be held in that position for 1 minute. Release and examine for compliance with 3.4.

4.4.1.2 Corrosion-resistance test. Clean both surfaces of shield. Then place one drop of copper-sulfate solution upon several places on each side of the shield. After remaining on the shield for 6 minutes, the solution shall be washed off with fresh water. If no copper has been deposited, the metal is corrosion resistant. The formulation of copper deposit within the 6 minutes indicates that the metal is not corrosion resistant, and shall be considered grounds for rejection. Copper-sulfate solution for this test shall be prepared as follows:

4 grams copper sulfate ($CuSO_4 5H_2O$).
18 grams sulfuric acid (H_2SO_4 concentrated).
90 grams distilled water (H_2O).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

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5.1.1 Level A. Twelve shields shall be packaged in an open-end, center-seam light-colored sulphite envelope. The envelope shall have a basis weight of 20 (17 x 22 - 500 sheets) and a bursting strength of 36 p.s.i.

51.2 Level B. Twelve shields shall be packaged in an open-end, Kraft envelope or in glassine envelopes in accordance with standard commercial practice.

5.1.3 Level C. The shields shall be packaged in accordance with the supplier's standard practice.

5.1.4 Intermediate packaging. Twelve (12) unit packages shall be intermediate packaged in boxes conforming to PPP-B-566, style II, or PPP-B-676. The boxes shall be closed and sealed in accordance with the applicable specification or appendix thereto.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The shields shall be packed in boxes conforming to PPP-B-636, class weather resistant V3c, RSC. The boxes shall be waterproof, sealed with tape, and reinforced with bands of reinforcing tape in accordance with the appendix to box specification.

5.2.2 Level B. Unless otherwise specified (see 6.2), 14 intermediate containers shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be sealed and closed in accordance with the appendix to the box specification.

5.2.3 Level C. The shields shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract or order the interior, intermediate, and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to any special markings required by the contract or order, the interior, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The metal erasing shield covered by this specification is intended for use by draftsmen in making erasures.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Whether special marking is required (see 5.3).

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Preparing activity:

GSA-FSS

CIVIL AGENCY INTEREST:

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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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